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Walking training system, operation method of the system, and non-transitory storage medium for the system

Abstract

A walking training system includes a leg orthosis fitted to a leg of a user, a pulling member coupled to the leg orthosis, a first drive mechanism configured to apply pull force to the pulling member, according to walking motion of the user, a holding device that holds the pulling member at a holding level, at a point between a coupling point of the leg orthosis and the first drive mechanism, and a second drive mechanism configured to change the holding level of the pulling member held by the holding device, according to the walking motion of the user.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2021-081118 filed on May 12, 2021, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

(2) The disclosure relates to a walking training system, an operation method of the system, and a non-transitory storage medium for the system.

2. Description of Related Art

(3) An example of a walking training apparatus is disclosed in Japanese Unexamined Patent Application Publication No. 2019-118706 (JP 2019-118706 A).

SUMMARY

(4) In the walking training apparatus, a leg orthosis may be fitted to a leg of a user, such as a rehabilitation patient. In the apparatus described in JP 2019-118706 A, the leg orthosis includes a

motor that generates driving force for assisting in extension motion and flexion motion of the knee joint. As the driving force of the motor is increased, the weight of the leg orthosis increases. As a result, an increased load is imposed on the trainee (user), which makes it difficult for the trainee to conduct training appropriately. Further, this may make wearing of the leg orthosis cumbersome and complicated.

(5) The disclosure provides a walking training system with which walking training is performed appropriately, and an operation method of the walking training system.

(6) A walking training system according to a first aspect of the disclosure includes a leg orthosis fitted to a leg of a user, at least one pulling member coupled to the leg orthosis, a first drive mechanism configured to apply pull force to the pulling member, according to a walking motion of the user, a holding device that holds the pulling member at a holding level, at a point between a coupling point of the leg orthosis and the first drive mechanism, and a second drive mechanism configured to change the holding level of the pulling member held by the holding device, according to the walking motion of the user.

(7) The leg orthosis may have a joint mechanism configured to pivot according to the walking motion of the user, and the first drive mechanism may be configured to pivot the joint mechanism, by pulling the pulling member.

(8) In the above walking training system, the pulling member may be coupled to the leg orthosis from ahead of the user, and the first drive mechanism may be configured to provide the pull force, by reeling in the pulling member.

(9) In the above walking training system, the holding device may have a suspended member that is suspended from above the user, and the second drive mechanism may be configured to raise the holding level, by reeling in the suspended member.

(10) The holding device may have a support member that extends upward, and the second drive mechanism may be configured to change the holding level, by extending and contracting the support member.

(11) In the above walking training system, the at least one pulling member may be a plurality of pulling members, and the plurality of pulling members may be connected to the leg orthosis from different directions.

(12) In the above walking training system, the pulling member may be coupled to the leg orthosis from one side of the user, and the leg orthosis may be provided with a mechanism configured to change a direction of the pull force of the pulling member.

(13) An operation method of a walking training system according to a second aspect of the disclosure is that of the walking training system including a leg orthosis fitted to a leg of a user, at least one pulling member coupled to the leg orthosis, a first drive mechanism configured to apply pull force to the pulling member, a holding device that holds the pulling member at a holding level, at a point between a coupling point of the leg orthosis and the first drive mechanism, and a second drive mechanism configured to change the holding level of the pulling member held by the holding device. The operation method includes detecting a walking motion of the user, and controlling the first drive mechanism and the second drive mechanism according to the walking motion of the user.

(14) In the above operation method, the leg orthosis may have a joint mechanism configured to pivot according to the walking motion of the user, and the first drive mechanism may pivot the joint mechanism, by pulling the pulling member.

(15) In the above operation method, the pulling member may be coupled to the leg orthosis from ahead of the user, and the first drive mechanism may provide the pull force, by reeling in the pulling member.

(16) In the above operation method, the holding device may have a suspended member that is suspended from above the user, and the second drive mechanism may raise the holding level, by reeling in the suspended member.

(17) In the above operation method, the holding device may have a support member that extends

upward, and the second drive mechanism may change the holding level, by extending and contracting the support member.

(18) In the above operation method, the at least one pulling member may be a plurality of pulling members, and the plurality of pulling members may be connected to the leg orthosis from different directions.

(19) In the above operation method, the pulling member may be coupled to the leg orthosis from one side of the user, and the leg orthosis may be provided with a mechanism configured to change a direction of the pull force of the pulling member.

(20) A third aspect of the disclosure relates to a non-transitory storage medium for a walking training system. The walking training system includes a leg orthosis fitted to a leg of a user, at least one pulling member coupled to the leg orthosis, a first drive mechanism configured to apply pull force to the pulling member, a holding device that holds the pulling member at a holding level, at a point between a coupling point of the leg orthosis and the first drive mechanism, and a second drive mechanism configured to change the holding level of the pulling member held by the holding device. The non-transitory storage medium stores instructions that are executable by one or more processors and that cause the one or more processors to perform functions comprising:

(21) detecting a walking motion of the user; and

(22) controlling the first drive mechanism and the second drive mechanism by the controller, according to the walking motion of the user.

(23) According to the disclosure, the walking training system with which walking training is performed appropriately, and the operation method of the system, are provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

(2) FIG. 1 is a schematic perspective view of a walking training system according to one embodiment;

(3) FIG. 2 is a side view showing the configuration of a leg orthosis;

(4) FIG. 3 is a front view showing the configuration of the leg orthosis;

(5) FIG. 4 is a view showing the internal configuration of the leg orthosis;

(6) FIG. 5 is a control block diagram showing a control system of the walking training system;

(7) FIG. 6 is a schematic view showing a main arrangement for driving the leg orthosis;

(8) FIG. 7 is a schematic view showing the trajectory of a knee joint;

(9) FIG. 8 is a schematic view showing the main configuration of a walking training system according to a first modified example;

(10) FIG. 9 is a schematic view showing the main configuration of a walking training system according to a second modified example; and

(11) FIG. 10 is a schematic view showing the main configuration of a walking training system according to a third modified example.

DETAILED DESCRIPTION OF EMBODIMENTS

(12) While the disclosure will be described through one embodiment of the disclosure, the disclosure will not be limited to the following embodiment. Also, not all of the configurations that will be described in the embodiment are necessarily essential as devices for solving the above problems.

First Embodiment

(13) A first embodiment will be described with reference to the drawings. FIG. 1 shows one

example of the configuration of a rehabilitation support system according to the first embodiment. The rehabilitation support system (walking training system **1**) according to this embodiment principally consists of a walking training apparatus **100** and a leg orthosis **120**. In the following description, an XYZ three-dimensional Cartesian coordinate system will be used as needed. In the coordinate system, the X direction is a direction in which a trainee **900** walks, and the Y direction is the lateral direction of the trainee **900**, while the Z direction is the vertical direction. Namely, the +X direction is forward of the trainee **900** who is undergoing walking training, and the -X direction is rearward of the trainee **900**. Similarly, the +Y direction is the left direction, and the -Y direction is the right direction, while the +Z direction is the upward direction, and the -Z direction is the downward direction, with respect to the trainee **900** in walking training.

(14) The walking training apparatus **100** is one specific example of a rehabilitation support apparatus that assists the trainee (user) **900** in rehabilitation. The walking training apparatus **100** is used by the trainee **900**, who is a hemiplegic patient suffering paralysis in one leg, when he/she performs walking training under the guidance of a trainer **901**. Here, the trainer **901** may be a therapist (physical therapist) or a physician, and may also be called “training guide”, “training helper”, or “training assistant” as the trainer **901** helps the trainee with training through guidance or assistance, for example.

(15) The walking training apparatus **100** mainly includes a control panel **133** mounted to a frame **130** that forms the overall framework, a treadmill **131** on which the trainee **900** walks, and the leg orthosis **120** fitted to an impaired leg as a paralyzed leg of the trainee **900**. In FIG. **1**, the leg orthosis **120** is fitted to the right leg of the trainee **900**.

(16) The frame **130** is erected on the treadmill **131** installed on a floor. The treadmill **131** rotates a ring-shaped belt **132** by means of a motor (not shown). The treadmill **131** promotes the trainee **900** to walk, and the trainee **900** who conducts walking training steps on the belt **132**, and tries walking motion in accordance with movement of the belt **132**. While the trainer **901** can walk together with the trainee **900** while standing on the belt **132** behind the trainee **900**, as shown in FIG. **1** by way of example, it is generally preferable that the trainer **901** is in a condition, like standing across the belt **132**, where he/she can easily help the trainee **900**.

(17) The frame **130** supports the control panel **133** that houses an overall controller **210** that controls motors and sensors, a monitor for training **138**, such as a liquid crystal panel, which presents the progress of training, etc. to the trainee **900**, and so forth. The frame **130** also supports an anterior pulling portion **135** located above and anterior to the head of the trainee **900**, a harness pulling portion **112** located above the head, and a posterior pulling portion **137** located above and posterior to the head. Also, the frame **130** includes handrails **130a** which allow the trainee **900** to grab or hold.

(18) The handrails **130a** are arranged on the right and left sides of the trainee **900**. Each of the handrails **130a** is disposed in parallel with the walking direction of the trainee **900**. The vertical and lateral positions of the handrails **130a** can be adjusted. Namely, the handrails **130a** may include a mechanism for changing their height and width. Further, the handrail **130a** may be constructed such that its inclination angle can be changed, by adjusting the height so that it varies between the front side and the rear side in the walking direction, for example. For example, the handrail **130a** may be formed with the inclination angle that gradually makes the handrail **130a** higher along the walking direction.

(19) Also, the handrail **130a** is provided with a handrail sensor **218** that detects a load received from the trainee **900**. For example, the handrail sensor **218** may be a load detection sheet of a resistance change detection type in which electrodes are arranged in a matrix. Also, the handrail sensor **218** may be a six-axis sensor as a composite of a three-axis acceleration sensor (x, y, z) and a three-axis gyro sensor (roll, pitch, yaw). It is, however, to be understood that the handrail sensor **218** may be of any type, and may be installed at any position.

(20) A camera **140** functions as an image pickup unit for observing the whole body of the trainee

900. The camera **140** is installed in the vicinity of the monitor for training **138**, to be opposed to the trainee. The camera **140** captures still images and moving images of the trainee **900** in training. The camera **140** includes a set of lens and image pickup device, which provides an angle of view large enough to shoot the whole body of the trainee **900**. The image pickup device may be a CMOS (Complementary Metal-Oxide-Semiconductor) image sensor, for example, which converts an optical image formed on an imaging plane into an image signal.

(21) Through coordinated movements of the anterior pulling portion **135** and the posterior pulling portion **137**, the load of the leg orthosis **120** is cancelled so as not to be a burden on the impaired leg, and, further, swing motion of the impaired leg is assisted according to the degree of settings.

(22) An anterior wire **134** is coupled at one end to a take-up mechanism of the anterior pulling portion **135**, and is coupled at the other end to the leg orthosis **120**. The take-up mechanism of the anterior pulling portion **135** has a motor (not shown), and reels in or reels out the anterior wire **134** in accordance with movement of the impaired leg, by turning on/off the motor. Similarly, a posterior wire **136** is coupled at one end to a take-up mechanism of the posterior pulling portion **137**, and is coupled at the other end to the leg orthosis **120**. The take-up mechanism of the posterior pulling portion **137** has a motor (not shown), and reels in or reels out the posterior wire **136** in accordance with movement of the impaired leg, by turning on/off the motor. Through the coordinated movements of the anterior pulling portion **135** and the posterior pulling portion **137**, the load of the leg orthosis **120** is cancelled, so as not to become a burden on the impaired leg, and the swing motion of the impaired leg is assisted according to the degree of settings.

(23) For example, the trainer **901**, as an operator, sets the level of assistance to a high level, for a trainee suffering a severe paralysis. When the level of assistance is set to a high level, the anterior pulling portion **135** reels in the anterior wire **134** with a relatively large force, in accordance with the swing timing of the impaired leg. As the training proceeds, and the trainee does not need assistance, the trainer **901** sets the level of assistance to a minimum. When the level of assistance is set to the minimum, the anterior pulling portion **135** reels in the anterior wire **134** with force that only cancels the self-weight of the leg orthosis **120**, in accordance with the swing timing of the impaired leg.

(24) The walking training apparatus **100** includes a fall-prevention harness device as a safety device, which principally consists of an orthosis **110**, a harness wire **111**, and a harness pulling portion **112**. The orthosis **110** is a belt that is wrapped around the abdomen of the trainee **900** and secured to the waist with a hook-and-loop fastener, for example. The orthosis **110** includes a connection hook **110a** to which one end of the harness wire **111** as a hanging member is coupled, and may also be called “hanger belt”. The trainee **900** wears the orthosis **110**, such that the connection hook **110a** is located on the back region.

(25) The harness wire **111** is coupled at one end to the connection hook **110a** of the orthosis **110**, and is coupled at the other end to a take-up mechanism of the harness pulling portion **112**. The take-up mechanism of the harness pulling portion **112** has a motor (not shown), and reels in or reels out the harness wire **111**, by turning on/off the motor. With this arrangement, when the trainee **900** is about to fall, the fall-prevention harness device reels in the harness wire **111** according to a command of the overall controller **210** that detects the movement of the trainee **900**, and causes the orthosis **110** to support the upper body of the trainee **900**, so as to prevent the trainee **900** from falling.

(26) The orthosis **110** includes a posture sensor **217** for detecting the posture of the trainee **900**. The posture sensor **217** is a combination of a gyro sensor and an acceleration sensor, for example, and outputs the inclination angle of the abdomen to which the orthosis **110** is fitted, relative to the direction of the gravity.

(27) The monitor for management **139**, which is attached to the frame **130**, is a display input device used mainly by the trainer **901** for monitoring and operation. The monitor for management **139** is a liquid crystal panel, for example, and a touch panel is provided on a surface of the monitor for

management **139**. The monitor for management **139** displays various menu items concerning training settings, various parameters for use in training, training results, and so forth. Also, an emergency stop button **232** is provided in the vicinity of the monitor for management **139**. When the trainer **901** pushes the emergency stop button **232**, the walking training apparatus **100** makes an emergency stop.

(28) The leg orthosis **120** is fitted to the impaired leg of the trainee **900**, and assist the trainee **900** in walking, by reducing a load of extension and flexion at the knee joint of the impaired leg. The leg orthosis **120** may include a sensor that measures a load at the sole, and so forth, and output various kinds of data concerning leg movements to the overall controller **210**. The leg orthosis **120** will be described in detail later.

(29) A wire **161** is connected to the leg orthosis **120**. The wire **161** is connected to a first drive mechanism **162**. The first drive mechanism **162** applies pull force to the wire **161**, according to the walking motion of the trainee **900**. The first drive mechanism **162** has a motor and a take-up mechanism for applying the pull force to the wire **161**. Namely, the first drive mechanism **162** reels in or reels out the wire **161**, by driving the motor. The wire **161** is attached at one end to the leg orthosis **120**, and is attached at the other end to the first drive mechanism **162**. The wire **161** is pulled around at a level around the knee height of the trainee **900**. The wire **161** is pulled around horizontally.

(30) The first drive mechanism **162** is located ahead of the trainee **900**. For example, the first drive mechanism **162** is housed in the control panel **133** located ahead of the treadmill **131**. The wire **161** is connected from ahead of the trainee **900** to the leg orthosis **120**. In operation, the first drive mechanism **162** can apply pull force to the leg orthosis **120**, by reeling in the wire **161**. The first drive mechanism **162** can weaken the pull force applied to the leg orthosis **120**, by reeling out the wire **161**. In this manner, a joint portion of the leg orthosis **120** is driven. In other words, the first drive mechanism **162** drives the wire **161**, so that the angle of the joint portion of the leg orthosis **120** becomes equal to a desired angle.

(31) A holding belt **171** is connected to the wire **161** at a point on the wire **161**. The holding belt **171** is located so as not to interfere with the trainee **900** in walking motion. For example, the holding belt **171** is located ahead of the trainee **900**. The holding belt **171** and the wire **161** are connected via a connecting portion **173**. For example, the holding belt **171** is wound around the wire **161**, so that the lower end of the holding belt **171** is attached to the wire **161**. The connecting portion **173** is a connection position at which the holding belt **171** and the wire **161** are connected to each other. The connection position of the holding belt **171** on the wire **161** may be fixed. The holding belt **171** provides a holding device that holds the wire **161** at a given level. The connecting portion **173** may have a coupling member, such as a ring or a hook. Namely, the holding belt **171** holds the wire **161** at the given level.

(32) Further, the holding belt **171** is connected to a second drive mechanism **172**. The second drive mechanism **172** is fixed to the frame **130**, at above the trainee **900**. The second drive mechanism **172** has a motor and a take-up mechanism for applying pull force to the holding belt **171**. The holding belt **171** is attached at the lower end to the connecting portion **173**, and is attached at the upper end to the second drive mechanism **172**. The holding belt **171** is suspended from the upper frame **130** down to the connecting portion **173**. The holding belt **171** is connected to the wire **161** at the connecting portion **173**.

(33) The second drive mechanism **172** reels in or reels out the holding belt **171**, so that the connecting portion **173** moves up and down. Accordingly, the level of the wire **161** is changed. The second drive mechanism **172** changes the holding level of the wire **161**. Namely, the second drive mechanism **172** drives the holding belt **171**, so that the level of the wire **161** becomes equal to a desired level. In this manner, the pulling direction of the wire **161** can be changed.

(34) The overall controller **210** generates rehabilitation data that may include configuration parameters concerning training settings, various kinds of data concerning leg movements generated

from various sensors as training results, and so forth. The rehabilitation data may include data indicating the trainer **901**, or his/her years of experience, level of proficiency, or the like, data indicating the symptoms, walking ability, degree of recovery, etc., of the trainee **900**, and various kinds of data transmitted from sensors, etc. The overall controller **210** generates a control signal for controlling each drive unit. The overall controller **210** controls each drive unit, based on detection signals from various sensors. Thus, it is possible to appropriately control the pull force and the take-up length, in accordance with the walking motion of the trainee **900**.

(35) Next, the leg orthosis **120** will be described, using FIG. 2 to FIG. 4. FIG. 2 to FIG. 4 show one example of the configuration of the leg orthosis **120**. FIG. 2 is a side view of the leg orthosis **120**, and FIG. 3 is a front view thereof. FIG. 4 is a cross-sectional view showing the internal configuration of a joint mechanism **123**.

(36) The leg orthosis **120** includes an upper-leg frame **121**, lower-leg frame **122**, and the joint mechanism **123**. The upper-leg frame **121** is fitted to the upper leg of the impaired leg of the trainee **900**. For example, the upper-leg frame **121** is fixed to the upper leg, using a band, hook-and-loop fastener, belt (not shown), or the like. Namely, the band, or the like, that extends from the upper-leg frame **121** is wrapped around the upper leg of the trainee **900**, so that the upper-leg frame **121** is placed along the upper leg. The upper-leg frame **121** is provided with an insertion opening **1211** for mounting the wire **161** in position. The insertion opening **1211** is a coupling point of the wire **161** with the leg orthosis **120**.

(37) The lower-leg frame **122** is fitted to the lower leg of the trainee **900**. For example, the lower-leg frame **122** is fixed to the lower leg, using a band, hook-and-loop fastener, belt (not shown), or the like. Namely, the band, or the like, that extends from the lower-leg frame **122** is wrapped around the lower leg of the trainee **900**, so that the lower-leg frame **122** is placed along the lower leg.

(38) The lower-leg frame **122** is connected to the upper-leg frame **121**, via the joint mechanism **123**. The joint mechanism **123** connects the lower-leg frame **122** with the upper-leg frame **121** such that the lower-leg frame **122** can freely pivot relative to the upper-leg frame **121**. The joint mechanism **123** corresponds to the knee joint of the impaired leg. The joint mechanism **123** pivotally connects the upper-leg frame **121** with the lower-leg frame **122**. The joint mechanism **123** may have a hinge or a bearing.

(39) The joint mechanism **123** pivots according to the motion of the knee joint of the trainee **900**. The joint mechanism **123** changes the angle formed by the upper-leg frame **121** and the lower-leg frame **122** in the direction of extension or flexion of the knee joint. Namely, the lower-leg frame **122** pivots about a joint axis **Ha** of the joint mechanism **123**, relative to the upper-leg frame **121**. The trainee wears the leg orthosis **120**, in a condition where the leg orthosis **120** is located at one side of the knee joint of the impaired leg. The upper-leg frame **121** and the lower-leg frame **122** pivot about the joint axis **Ha**. Thus, during walking training, the leg orthosis **120** can assist movement of the knee joint. In this connection, movable ranges of the upper-leg frame **121** and the lower-leg frame **122** may be mechanically restricted by a stopper, for example.

(40) As shown in FIG. 3, the joint mechanism **123** is provided with a biasing member **1235**. The biasing member **1235** is a torsion spring, for example, and generates biasing force between the upper-leg frame **121** and the lower-leg frame **122**, so that the knee joint extends. Namely, the biasing member **1235** generates biasing force, so that the upper-leg frame **121** and the lower-leg frame **122** extend in parallel with each other.

(41) As shown in FIG. 2 and FIG. 4, the wire **161** is mounted in the joint mechanism **123**. For example, the wire **161** is inserted through the insertion opening **1211** of the upper-leg frame **121**, into the joint mechanism **123**. A distal end of the wire **161** is fixed to a stationary end **1237** of the joint mechanism **123**. Also, at the insertion opening **1211**, an outer **1610** of the wire **161** is fixed to the upper-leg frame **121**.

(42) The first drive mechanism **162** pulls the wire **161**, so that the joint mechanism **123** pivots in

such a direction as to flex the knee joint. More specifically, as shown in FIG. 2, the wire **161** is pulled in a direction indicated by a straight arrow, so that the length of the wire **161** from the stationary end **1237** to the insertion opening **1211** is reduced. As a result, the stationary end **1237** gets closer to the insertion opening **1211**, so that the joint mechanism **123** pivots in the direction to flex the knee joint. Also, when the wire **161** stops being pulled, the joint mechanism **123** pivots in the direction of extension, under the bias force of the biasing member **1235**. The leg orthosis **120** can assist extension and flexion of the knee joint, in accordance with the walking motion of the trainee **900**.

(43) Thus, it is possible to change the joint angle of the joint mechanism **123**, by using the wire **161** from the outside of the leg orthosis **120**. Namely, when the pull force of the wire **161** is increased, the joint mechanism **123** is driven in the direction to flex the knee joint. Also, when the pull force of the wire **161** is reduced, the joint mechanism **123** is driven, in the direction to extend the knee joint under the bias force of the biasing member **1235**. Thus, the overall controller **210** controls the pull force applied from the first drive mechanism **162** shown in FIG. 1 to the wire **161**, according to movement of the knee joint of the trainee **900**.

(44) Thus, the movement of the knee joint can be supported, by use of the pull force of the wire **161** connected from the outside to the leg orthosis **120**. In other words, the motor and actuator for applying drive force to the leg orthosis **120** can be placed outside the leg orthosis **120**. With this arrangement, electric equipment, such as a motor, can be placed outside the leg orthosis. For example, the leg orthosis **120** does not require a motor, and a power supply for driving the motor. Thus, the weight and size of the leg orthosis **120** can be reduced. The joint mechanism **123** may be provided with an angle sensor that detects the angle formed by the upper-leg frame **121** and the lower-leg frame **122**.

(45) Referring next to FIG. 5, an example of the system configuration of the walking training apparatus **100** will be described. FIG. 5 is a block diagram showing the example of the system configuration of the walking training apparatus **100**. As shown in FIG. 5, the walking training apparatus **100** may include the overall controller **210**, a treadmill drive unit **211**, an operation accepting unit **212**, a display controller **213**, and a pulling drive unit **214**. The walking training apparatus **100** may also include a harness drive unit **215**, image processing unit **216**, posture sensor **217**, handrail sensor **218**, communication connection interface (IF), and input-output unit **231**. The walking training apparatus **100** includes a joint drive unit **221**, load sensor **222**, angle sensor **223**, and auxiliary drive unit **224**.

(46) The overall controller **210** is a Micro Processing Unit (MPU), for example, and executes control programs read from a system memory, to perform control on the apparatus as a whole.

(47) The treadmill drive unit **211** includes a motor that rotates the belt **132**, and a drive circuit for the motor. The overall controller **210** sends a drive signal to the treadmill drive unit **211**, to control rotation of the belt **132**. The overall controller **210** adjusts the rotational speed of the belt **132**, according to the walking speed set by the trainer **901**, for example.

(48) The operation accepting unit **212** accepts input operation from the trainer **901**, and sends an operation signal to the overall controller **210**. The trainer **901** operates operation buttons provided on the apparatus, a touch panel superposed on the monitor for management **139**, accompanying remote controller, etc. which constitute the operation accepting unit **212**. Through the operation, it is possible to turn on/off the power supply, give a command to start training, enter numerical values concerning settings, and select menu items. The operation accepting unit **212** may also accept input operation from the trainee **900**.

(49) The display controller **213** receives a display signal from the overall controller **210**, generates a display image, and displays the image on the monitor for training **138** or monitor for management **139**. The display controller **213** generates an image indicating the progress of training, or a real-time picture taken by the camera **140**, according to the display signal.

(50) The pulling drive unit **214** includes a motor for pulling the anterior wire **134** and its drive

circuit, which constitute the anterior pulling portion **135**, and a motor for pulling the posterior wire **136** and its drive circuit, which constitute the posterior pulling portion **137**. The overall controller **210** sends a drive signal to the pulling drive unit **214**, to control winding of the anterior wire **134** and winding of the posterior wire **136**, respectively. Also, the overall controller **210** controls the pull force of each wire, as well as the winding operation, by controlling drive torque of the motor. The overall controller **210** identifies the time of switching of the impaired leg from a stance phase to a swing phase, from the detection result of the load sensor **222**, for example, and increases or reduces the pull force of each wire in synchronization with the switching time, to thus assist swinging motion of the impaired leg.

(51) The harness drive unit **215** includes a motor for pulling the harness wire **111** and its drive circuit, which constitute the harness pulling portion **112**. The overall controller **210** sends a drive signal to the harness drive unit **215**, to control winding of the harness wire **111** and the pull force of the harness wire **111**. When the overall controller **210** predicts a fall of the trainee **900**, for example, it reels in the harness wire **111** by a given amount, to prevent the fall of the trainee.

(52) The image processing unit **216** is connected to the camera **140**, and can receive an image signal from the camera **140**. The image processing unit **216** receives the image signal from the camera **140**, according to a command from the overall controller **210**, and generates image data by processing the received image signal. Also, the image processing unit **216** performs image processing on the image signal received from the camera **140**, according to a command from the overall controller **210**, to carry out a particular image analysis. For example, the image processing unit **216** detects a foot position (stance position) of the impaired leg that contacts the treadmill **131**, through the image analysis. More specifically, the image processing unit **216** extracts an image area around a tip of the foot or toe, for example, and computes the stance position, by analyzing an identification marker depicted on the belt **132** that overlaps the tip portion.

(53) The posture sensor **217** detects the inclination angle of the abdomen region of the trainee **900** relative to the direction of the gravity, as described above, and sends a detection signal to the overall controller **210**. The overall controller **210** computes the posture of the trainee **900**, more specifically, the inclination angle of the trunk, using the detection signal from the posture sensor **217**. The overall controller **210** and the posture sensor **217** may be connected by wire, or may be wirelessly connected via near-field communication.

(54) The handrail sensor **218** detects a load applied to the handrail **130a**. Namely, the load corresponding a part of the body weight of the trainee **900** that cannot be supported by his/her both legs is applied to the handrail **130a**. The handrail sensor **218** detects the load, and sends a detection signal to the overall controller **210**.

(55) The overall controller **210** also plays a role of a function executing unit that performs various computations concerning control, and controls. The overall controller **210** controls each drive unit, using data obtained from various sensors.

(56) The joint drive unit **221** includes a drive control circuit of the first drive mechanism **162** (see FIG. **1**). The joint drive unit **221** generates a drive signal of a take-up motor, according to a control signal from the overall controller **210**. The joint drive unit **221** sends the drive signal to the first drive mechanism **162**, so that the first drive mechanism **162** reels in or reels out the wire **161**. As a result, the upper-leg frame **121** and the lower-leg frame **122** are forced to open relative to each other about the joint axis H_a , or forced to close. With the movements, the extending motion and flexing motion of the knee are assisted, and the knee is prevented from buckling. The joint drive unit **221** drives the first drive mechanism **162**, according to the walking motion. It is possible to control the joint mechanism **123** of the leg orthosis **120**, according to the knee joint movement of the trainee **900**.

(57) The load sensor **222** detects the magnitude and distribution of vertical loads received by the sole of the foot of the trainee **900** as described above, and sends a detection signal to the overall controller **210**. The load sensor **222** may be provided on the treadmill **131**. When the leg orthosis

120 has a sole frame attached to the sole of the trainee **900**, the load sensor **222** may be provided on the sole frame. Also, the load sensor **222** may be attached to a shoe of the trainee **900**. The overall controller **210** determines the swing or stance phase, or estimates switching of the phases, based on the detection signal from the load sensor **222**.

(58) The angle sensor **223** detects the angle formed by the upper-leg frame **121** and the lower-leg frame **122** about the joint axis Ha as described above, and sends a detection signal to the overall controller **210**. The overall controller **210** receives the detection signal, and computes the opening angle of the knee joint. The angle sensor **223** may be provided on the leg orthosis **120**, or may be provided on a place other than the leg orthosis **120**. For example, a cameral that shoots the trainee **900** from one side thereof may be used as the angle sensor **223**. In this case, the image processing unit **216**, for example, can perform image analysis, and obtain the opening angle of the knee joint.

(59) The auxiliary drive unit **224** includes a drive control circuit of the second drive mechanism **172** (see FIG. 1). The auxiliary drive unit **224** generates a drive signal of a take-up motor, according to a control signal from the overall controller **210**. The auxiliary drive unit **224** sends the drive signal to the second drive mechanism **172**, so that the second drive mechanism **172** reels in or reels out the holding belt **171**. In this manner, the second drive mechanism **172** can control the level of the wire **161**.

(60) The auxiliary drive unit **224** drives the second drive mechanism **172**, according to the walking motion. For example, the knee height of the trainee **900** varies according to the walking motion. The auxiliary drive unit **224** changes the level of the wire **161** so as to follow the variation in the knee height. In other words, the auxiliary drive unit **224** raises the holding level of the wire **161**, at the time when the knee height of the trainee **900** rises. To the contrary, the auxiliary drive unit **224** lowers the holding level of the wire **161**, at the time when the knee height of the trainee **900** becomes lower. Thus, the auxiliary drive unit **224** can control the level of the wire **161**, according to the trajectory of the knee joint of the trainee **900**.

(61) The overall controller **210** detects the walking motion of the trainee **900**. The overall controller **210** detects the walking motion with various sensors, such as the angle sensor **223**, load sensor **222**, and posture sensor **217**. Then, the overall controller **210** controls the first drive mechanism **162** and the second drive mechanism **172**, according to the walking motion of the trainee **900**. The overall controller **210** detects the time of switching between the swing phase and the stance phase in a walking motion, take-off time, landing time, etc., according to detection signals from various sensors. The sensors for detecting the phase and timing of walking motion are not limited to any particular ones, since known methods can be used.

(62) The overall controller **210** controls the first drive mechanism **162** and the second drive mechanism **172**, according to the walking motion. The overall controller **210** generates a control signal for each drive unit, according to the phase of the walking motion. The overall controller **210** outputs control signals to the auxiliary drive unit **224**, joint drive unit **221**, pulling drive unit **214**, and harness drive unit **215**. The auxiliary drive unit **224**, joint drive unit **221**, pulling drive unit **214**, and harness drive unit **215** drive the corresponding motors, etc. For example, drive patterns corresponding to the respective phases and timing of walking motion may be registered. Then, the overall controller **210** generates control signals corresponding to the drive patterns. The drive pattern may also reflect body information, such as the height, weight, leg length, upper-leg length, and lower-leg length, of the trainee **900**.

(63) Referring to FIG. 6, driving of the leg orthosis **120** will be described. FIG. 6 is a side view schematically showing the walking training system. In FIG. 6, the configuration shown in FIG. 1 is simplified as appropriate. More specifically, FIG. 6 indicates the configuration related to the leg orthosis **120** and its driving. For example, in FIG. 6, the harness wire **111**, harness pulling portion **112**, handrails **130a**, anterior wire **134**, anterior pulling portion **135**, posterior wire **136**, posterior pulling portion **137**, etc., are not illustrated.

(64) The leg orthosis **120** is fitted to the right leg of the trainee **900**. More specifically, the upper-

leg frame **121** and the lower-leg frame **122** are attached to the upper leg and lower leg of the right leg, respectively, with bands **128**. Namely, the band **128** is wrapped around the upper leg, so that the upper-leg frame **121** is fixed along the upper leg. Similarly, the band **128** is wrapped around the lower leg, so that the lower-leg frame **122** is fixed along the lower leg. The upper-leg frame **121** and the lower-leg frame **122** are pivotally connected via the joint mechanism **123**.

(65) Further, a sole frame **126** is attached to the lower end of the lower-leg frame **122**. The trainee **900** places the right leg on the sole frame **126**. The sole frame **126** is in contact with the belt **132** of the treadmill **131**. The sole frame **126** may be provided with the load sensor **222** as described above. Needless to say, when the impaired leg of the trainee **900** is the left leg, the trainee **900** may wear the leg orthosis **120** on the left leg.

(66) The wire **161** is connected to the joint mechanism **123**. The wire **161** connects the first drive mechanism **162** with the joint mechanism **123**. The first drive mechanism **162** reels in the wire **161**, so that the pull force is applied to the wire **161**. As a result, the angle of the joint mechanism **123** changes. Thus, the leg orthosis **120** can assist movement of the knee joint, according to the walking motion.

(67) The first drive mechanism **162** is located outside the leg orthosis **120**. The first drive mechanism **162** can apply force for assisting movement of the knee joint, from the outside of the leg orthosis **120**. Thus, the leg orthosis **120** does not need to incorporate a motor, power supply, etc. Thus, the weight and size of the leg orthosis **120** can be reduced.

(68) Further, the holding belt **171** can be located so as not to interfere with the trainee **900** in walking training. For example, the holding belt **171** is located ahead of the trainee **900**. The first drive mechanism **162** and the second drive mechanism **172** may also be located so as not to interfere with the trainee **900**. For example, the second drive mechanism **172** is located above the trainee **900**, and the first drive mechanism **162** is located ahead of the trainee **900** and the holding belt **171**. Thus, the trainee **900** can conduct walking training, without interfering with each drive mechanism, etc.

(69) The wire **161** is made thicker, so as to increase drive torque of the joint mechanism **123**. For example, the diameter of the wire **161** is made equal to 15 mm, so that sufficient pull force is applied to the joint mechanism **123**. As the weight of the wire **161** increases, the wire **161** becomes less likely to bend. When the knee joint of the trainee **900** moves, the wire **161** may give resistance that impedes movement of the trainee **900**. Since the second drive mechanism **172** changes the holding level of the holding belt **171**, the wire **161** is prevented from impeding movement of the trainee **900**. Also, there is a possibility of an abrupt bend in the wire **161**, from the joint mechanism **123** toward the ground. With the holding belt **171** keeping the level of the wire **161**, as in this embodiment, the abrupt bend of the wire **161** can be prevented.

(70) When the first drive mechanism **162** is located ahead of the treadmill **131**, the length of the wire **161** is increased. In this case, the wire **161** may be brought into contact with the ground, owing to deflection of the wire **161** due to its self-weight. However, the wire **161** can be prevented from contacting with the ground, since the holding belt **171** keeps the level of the wire **161**.

(71) It is preferable to keep a predetermined distance from a coupling position of the wire **161** and the leg orthosis **120**, to the joint mechanism **123**, so as to prevent unnecessary pull force from being applied to the joint mechanism **123**. The wire **161** is preferably pulled around at right angles with respect to the upper leg. In this case, the wire **161** and the connecting portion **173** are preferably designed with a certain degree of freedom in the lateral direction. Namely, a mechanism for moving the wire **161** and the connecting portion **173** in the lateral direction according to the walking motion may be provided.

(72) It is preferable that the wire **161** is connected to the leg orthosis **120**, from the front side of the trainee **900**. With this arrangement, the wire **161** is prevented from interfering with the trainer **901** and other configurations. With this arrangement, the trainer **901** can help the trainee **900** from behind. Also, even when the handrail **130a** is present at one side of the trainee **900**, the wire **161**

can be pulled around without interfering with the handrail **130a**.

(73) By pulling the wire **161** around from the front side, the pull-around direction of the wire **161** can be made in parallel with or close to the horizontal direction. With the wire **161** thus pulled around horizontally, the wire **161** can be prevented from interfering with the arm of the trainee **900**. It is preferable to apply pull force in a direction perpendicular to the upper leg. The wire **161** is pulled around from the front in the horizontal direction, to be connected to the leg orthosis **120**. The first drive mechanism **162** is preferably located at around the knee height in front of and ahead of the trainee **900**.

(74) FIG. 7 is a side view schematically showing the result obtained by calculating the trajectory of the knee joint of the trainee **900**. In FIG. 7, the distance from the hip joint to the knee joint of the trainee is assumed to be 50 cm. Accordingly, the trajectory T of the knee joint is a circular arc having a center O, and a radius of 50 cm. Namely, in walking motion, the position of the knee joint relative to the hip joint changes along a track of the arc having the radius of 50 cm.

(75) The minimum bending radius of the wire **161** is assumed to be 100 cm, and the remaining portion extends horizontally. Thus, the wire **161** is divided into a straight portion **161a** and an arc portion **161b**. The arc portion **161b** is a portion closer to the distal end of the wire **161**, namely, closer to the leg orthosis **120**. The straight portion **161a** is provided on the first drive mechanism **162** side (not shown in FIG. 7), and extends in parallel with the X direction.

(76) At this time, it is desirable that the wire **161** is connected at right angles with the upper leg; therefore, the trajectory T of the knee joint is connected to the arc portion **161b** of the wire **161** with a tangent line. The arc portion **161b** and the trajectory T are smoothly connected, at the coupling portion C of the wire **161** and the leg orthosis **120**. The trajectory of the optimum holding level of the connecting portion **173** in this case is indicated as “trajectory H”. Thus, the second drive mechanism **172** may change the level of the connecting portion **173**, so that it follows the trajectory H.

(77) In the above description, the wire **161** is used, as the pulling member that applies pull force to the joint mechanism **123** of the leg orthosis **120**. The pull member is not limited to the wire **161**, but may be a rope, string, band, belt, etc. The pull member may be formed of a material, such as metal, resin, or fiber. The holding belt **171** can keep the holding level, even if the pulling member is made of a material that deflects due to the self-weight, for example.

(78) Similarly, the holding device that holds the wire **161** at a given level is not limited to the holding belt **171**. The holding device is not limited to the belt, but may be a wire, rope, string, or band. The holding device may be made of a material, such as metal, resin, or fiber. The holding device is preferably a suspended member that is suspended from above. With this arrangement, the pull force that pulls the suspended member upward can be applied to the suspended member, by means of a motor and a take-up mechanism, for example. The second drive mechanism **172** reels in the suspended member, thereby to apply pull force to pull the suspended member upward. Thus, the holding level can be controlled with a simple arrangement. Further, the XY position of the holding belt **171** may be changed, according to the walking motion.

(79) Also, the joint assisted by the leg orthosis **120** is not limited to the knee joint. For example, the leg orthosis **120** may assist the hip joint or the ankle joint in joint motion. The leg orthosis **120** may assist in two or more joint motions. In this case, two or more wires **161** and two or more drive mechanisms may be provided, depending on the number of the joints to be assisted.

First Modified Example

(80) The system configuration according to a first modified example will be described with reference to FIG. 8. In the first modified example, the holding device that holds the wire **161** at a holding level is different from that of the above embodiment. Here, a support member **175** holds the wire **161** at a given level.

(81) The support member **175** extends upward from the floor. The support member **175** is able to extend and contract. The upper end of the support member **175** provides a connecting portion **173**

that connects with the wire **161**. For example, the wire **161** is fixed to the upper face of the support member **175**. The second drive mechanism **172** has a motor and a cylinder for extending and contracting the support member **175**. The second drive mechanism **172** extends or contracts the support member **175**, so that the level of the connecting portion **173** changes. Namely, when the support member **175** extends or contracts, the holding level of the wire **161** changes. With this configuration, too, substantially the same effects as those of the first embodiment can be obtained.

Second Modified Example

(82) The system configuration according to a second modified example will be described with reference to FIG. **9**. In the second modified example, a wire **166**, third drive mechanism **182**, connecting portion **183**, holding belt **184**, and fourth drive mechanism **185** are provided, in addition to the configuration of FIG. **6**. Namely, two wires **161**, **166** are connected to the leg orthosis **120**. The arrangement associated with the wire **161** is similar to that of the first embodiment, and will not be described in detail. Namely, the first drive mechanism **162** applies pull force to the wire **161**, so that the joint mechanism **123** moves in a direction to flex the knee joint.

(83) The wire **166** is connected to the leg orthosis **120** from the rear side. The wire **166** is connected to the third drive mechanism **182**. The third drive mechanism **182** is located behind the treadmill **131**. The third drive mechanism **182** includes a motor and a take-up mechanism, like the first drive mechanism **162**.

(84) The third drive mechanism **182** can apply pull force to the leg orthosis **120**, by reeling in the wire **166**. The third drive mechanism **182** can weaken the pull force applied to the leg orthosis **120**, by reeling out the wire **166**. In this manner, the joint portion of the leg orthosis **120** is driven. In other words, the third drive mechanism **182** drives the wire **166**, so that the angle of the joint portion of the leg orthosis **120** becomes equal to a desired angle.

(85) In the connecting portion **183**, the wire **166** is connected to the holding belt **184**. Namely, the holding belt **184** holds the wire **166** via the connecting portion **183**. The holding belt **184** holds the wire **166** so that the level of the wire **166** becomes equal to a given level.

(86) The holding belt **184** is attached to the fourth drive mechanism **185**. The fourth drive mechanism **185** has a motor and a take-up mechanism, like the second drive mechanism **172**. Thus, the fourth drive mechanism **185** can change the holding level of the wire **166**, by reeling in or reeling out the holding belt **184**.

(87) The two or more wires **161**, **166** are connected to the leg orthosis **120** from different directions. The pull force provided by the first drive mechanism **162** and the pull force provided by the third drive mechanism **182** can be applied in different directions. The pull force is applied forward from the wire **161** to the leg orthosis **120**, and the pull force is applied backward from the wire **166** to the leg orthosis **120**. With the two wires **161**, **166** thus provided, the joint mechanism **123** can be pushed and pulled. With this arrangement, the pull force can be applied both forward and backward. Thus, two or more wires are connected to the leg orthosis **120**, so that the pull force can be applied in two or more directions.

(88) Furthermore, the drive direction of the joint mechanism **123** can be varied, with the pull force provided by the first drive mechanism **162** and the pull force provided by the third drive mechanism **182**. For example, the joint mechanism **123** moves in the direction to extend the knee joint, due to the pull force of the third drive mechanism **182**. The joint mechanism **123** moves in the direction to flex the knee joint, due to the pull force of the first drive mechanism **162**. Thus, the angle of the knee joint can be quickly made equal to a desired angle; therefore, the motion of the knee joint can be supported more appropriately.

Third Modified Example

(89) In a third modified example, the wire **161** is pulled around from one side of the trainee **900**, as shown in FIG. **10**. More specifically, the wire **161** is connected to the leg orthosis **120** from the -Y side. A distal end of the wire **161** is connected to the leg orthosis **120**, and the other end is connected to the first drive mechanism **162**. Then, the first drive mechanism **162** reels in the wire

161, so that pull force is applied to the leg orthosis **120**. In this manner, the joint mechanism **123** is driven.

(90) The holding belt **171** holds the wire **161**. The holding belt **171** is connected to the wire **161** at a point on the wire **161**. The holding belt **171** and the wire **161** are connected via the connecting portion **173**. With this arrangement, the second drive mechanism **172** can raise and lower the holding level of the wire **161**, according to the walking motion.

(91) Further, the leg orthosis **120** is provided with a shifting mechanism **129** that changes the direction of pull force. The shifting mechanism **129** has a pulley, for example. Then, the shifting mechanism **129** changes the direction of the wire **161** provided along the lateral direction. More specifically, when the direction of the wire **161** is changed by 90°, the pull force can be applied from ahead of the leg orthosis **120**. In this manner, the first drive mechanism **162** can pull the wire **161** in the lateral direction.

(92) The operation method of the walking training system can be implemented by a computer program or hardware. The overall controller **210** includes a memory that stores the program, and a processor, etc. that executes the program. The overall controller **210** can implement the operation method of the walking training system **1** according to this embodiment, by executing the program.

(93) The program as described above may be stored and supplied to a computer, using any of various types of non-transitory computer-readable media. The non-transitory computer-readable media include various types of tangible recording media. Examples of the non-transitory computer-readable media include a magnetic recording media (e.g., a flexible disc, magnetic tape, hard disc drive), magneto-optical recording medium (e.g., magneto-optical disc), CD-ROM (Read Only Memory), CD-R, CD-R/W, and semiconductor memory (e.g., mask ROM, PROM (Programmable ROM), EPROM (Erasable PROM), flash ROM, RAM (Random Access Memory)). Also, the program may be supplied to the computer, via any of various types of transitory computer-readable media. Example of the transitory computer-readable media include an electric signal, optical signal, and electromagnetic wave. The transitory computer-readable medium can supply the program to the computer, via a wire communication channel, such as an electric wire or an optical fiber, or a wireless communication channel.

(94) The disclosure is not limited to the above embodiment, but may be changed as appropriate, without departing from the principle.

Claims

1. A walking training system comprising: a leg orthosis configured to fit to a leg of a user; at least one pulling member coupled to the leg orthosis; a first drive mechanism configured to apply pull force to the at least one pulling member, according to a walking motion of the user; a holding device that holds the at least one pulling member at a holding level, the holding device attached to the at least one pulling member at a location of the at least one pulling member that is between (a) a coupling point at which the at least one pulling member is attached to the leg orthosis and (b) the first drive mechanism; a second drive mechanism configured to change the holding level of the at least one pulling member held by the holding device, according to the walking motion of the user; and a controller configured to control the first drive mechanism and the second drive mechanism, wherein: the leg orthosis includes (i) an upper frame configured to be attached to the leg of the user above a knee of the user, (ii) a lower frame configured to be attached to the leg of the user below the knee of the user, and (iii) a joint mechanism that pivotally connects the upper frame to the lower frame and is configured to pivot according to the walking motion of the user, the at least one pulling member is connected to the joint mechanism, the first drive mechanism is configured to pivot the joint mechanism by pulling the at least one pulling member under control of the controller, and the second drive mechanism is configured to change the holding level of the at least one pulling member to follow a height of the joint mechanism which changes as the joint

mechanism pivots, so that the holding level rises as the joint mechanism rises and the holding level lowers as the joint mechanism lowers due to pivoting of the joint mechanism.

2. The walking training system according to claim 1, wherein: the at least one pulling member is coupled to the leg orthosis from ahead of the user; and the first drive mechanism is configured to provide the pull force, by reeling in the at least one pulling member under control of the controller.

3. The walking training system according to claim 1, wherein: the holding device has a suspended member that is suspended from above the user; and the second drive mechanism is configured to raise the holding level, by reeling in the suspended member under control of the controller.

4. The walking training system according to claim 1, wherein: the holding device has a support member that extends upward; and the second drive mechanism is configured to change the holding level, by extending and contracting the support member under control of the controller.

5. The walking training system according to claim 1, wherein the at least one pulling member is a plurality of pulling members, and the plurality of pulling members is connected to the leg orthosis from different directions.

6. The walking training system according to claim 1, wherein: the at least one pulling member is coupled to the leg orthosis from one side of the user; and the leg orthosis is provided with a mechanism configured to change a direction of the pull force of the pulling member.

7. The walking training system according to claim 1, wherein: the first drive mechanism is configured to drive the at least one pulling member so that an angle of the joint mechanism becomes equal to a determined angle.

8. The walking training system according to claim 1, further comprising: a harness configured to be attached to the user separately from the orthosis, the harness including a posture sensor configured to detect a posture of the user, wherein the controller controls the first drive mechanism and the second drive mechanism based on a detection result of the posture sensor.

9. The walking training system according to claim 1, wherein: the at least one pulling member is inserted into the joint mechanism.

10. An operation method of a walking training system including (i) a leg orthosis configured to fit to a leg of a user, (ii) at least one pulling member coupled to the leg orthosis, (iii) a first drive mechanism configured to apply pull force to the at least one pulling member, (iv) a holding device that holds the at least one pulling member at a holding level, the holding device attached to the at least one pulling member at a location of the at least one pulling member that is between (a) a coupling point at which the at least one pulling member is attached to the leg orthosis and (b) the first drive mechanism, (v) a second drive mechanism configured to change the holding level of the at least one pulling member held by the holding device, and (vi) a controller configured to control the first drive mechanism and the second drive mechanism, (A) the leg orthosis including (1) an upper frame configured to be attached to the leg of the user above a knee of the user, (2) a lower frame configured to be attached to the leg of the user below the knee of the user, and (3) a joint mechanism that pivotally connects the upper frame to the lower frame and is configured to pivot according to a walking motion of the user, and (B) the at least one pulling member is connected to the joint mechanism, the operation method comprising: detecting the walking motion of the user; and controlling the first drive mechanism and the second drive mechanism by the controller, according to the walking motion of the user, wherein the controller controls the first drive mechanism to pivot the joint mechanism by pulling the at least one pulling member, and the controller controls the second drive mechanism to change the holding level of the at least one pulling member to follow a height of the joint mechanism which changes as the joint mechanism pivots, so that the holding level rises as the joint mechanism rises and the holding level lowers as the joint mechanism lowers due to pivoting of the joint mechanism.

11. A non-transitory storage medium for a walking training system including (i) a leg orthosis configured to fit to a leg of a user, (ii) at least one pulling member coupled to the leg orthosis, (iii) a first drive mechanism configured to apply pull force to the at least one pulling member, (iv) a

holding device that holds the at least one pulling member at a holding level, the holding device attached to the at least one pulling member at a location of the at least one pulling member that is between (a) a coupling point at which the at least one pulling member is attached to the leg orthosis and (b) the first drive mechanism, and (v) a second drive mechanism configured to change the holding level of the at least one pulling member held by the holding device, (A) the leg orthosis including (1) an upper frame configured to be attached to the leg of the user above a knee of the user, (2) a lower frame configured to be attached to the leg of the user below the knee of the user, and (3) a joint mechanism that pivotally connects the upper frame to the lower frame and is configured to pivot according to a walking motion of the user, and (B) the at least one pulling member is connected to the joint mechanism, wherein the non-transitory storage medium stores instructions that are executable by one or more processors and that cause the one or more processors to perform functions comprising: detecting the walking motion of the user; and controlling the first drive mechanism and the second drive mechanism by the one or more processors, according to the walking motion of the user that has been detected, wherein the one or more processors control the first drive mechanism to pivot the joint mechanism by pulling the at least one pulling member, and the one or more processors control the second drive mechanism to change the holding level of the at least one pulling member to follow a height of the joint mechanism which changes as the joint mechanism pivots, so that the holding level rises as the joint mechanism rises and the holding level lowers as the joint mechanism lowers due to pivoting of the joint mechanism.
